

Lecture # 12

MM 318: Multicomponent Alloys

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Boltzmann-Transport Equation

Reference:

Harald Ibach, Hans Luth, Solid-state physics : An Introduction to principle of materials science

Why Boltzmann Transport Eqn. (BTE)?

- Originally derived for a dilute gas of non-interacting particles
- Extended to the simulation of electron and phonon transport
- Can include many other important effects:
 - electron band structure
 - phonon dispersion
 - Electro-thermal transport

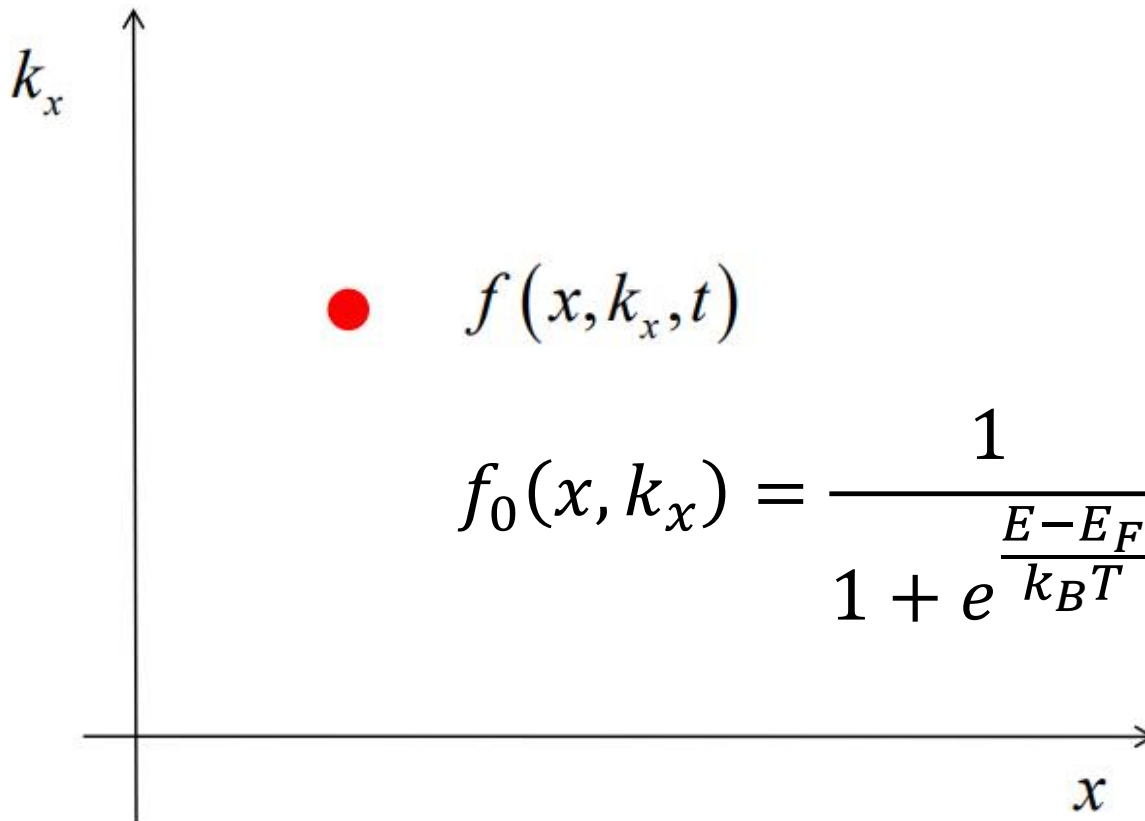
Distribution function

- Distribution function $f_T(r, k, t)$ represents the probability for a particle to occupy position r with momentum k at time t .
- Distribution function f_T contains all the information about the transport in the system.
- From f_T ; we can obtain average quantities like current, mobility, mean-free-path, etc.
- It is 7-D in general: 3-D spatial (r) + 3-D momentum (k) + time (t) dependence.
- In 1-D materials like CNTs and nanowires, space and momentum are 1-D, so f_T is 3-D altogether.

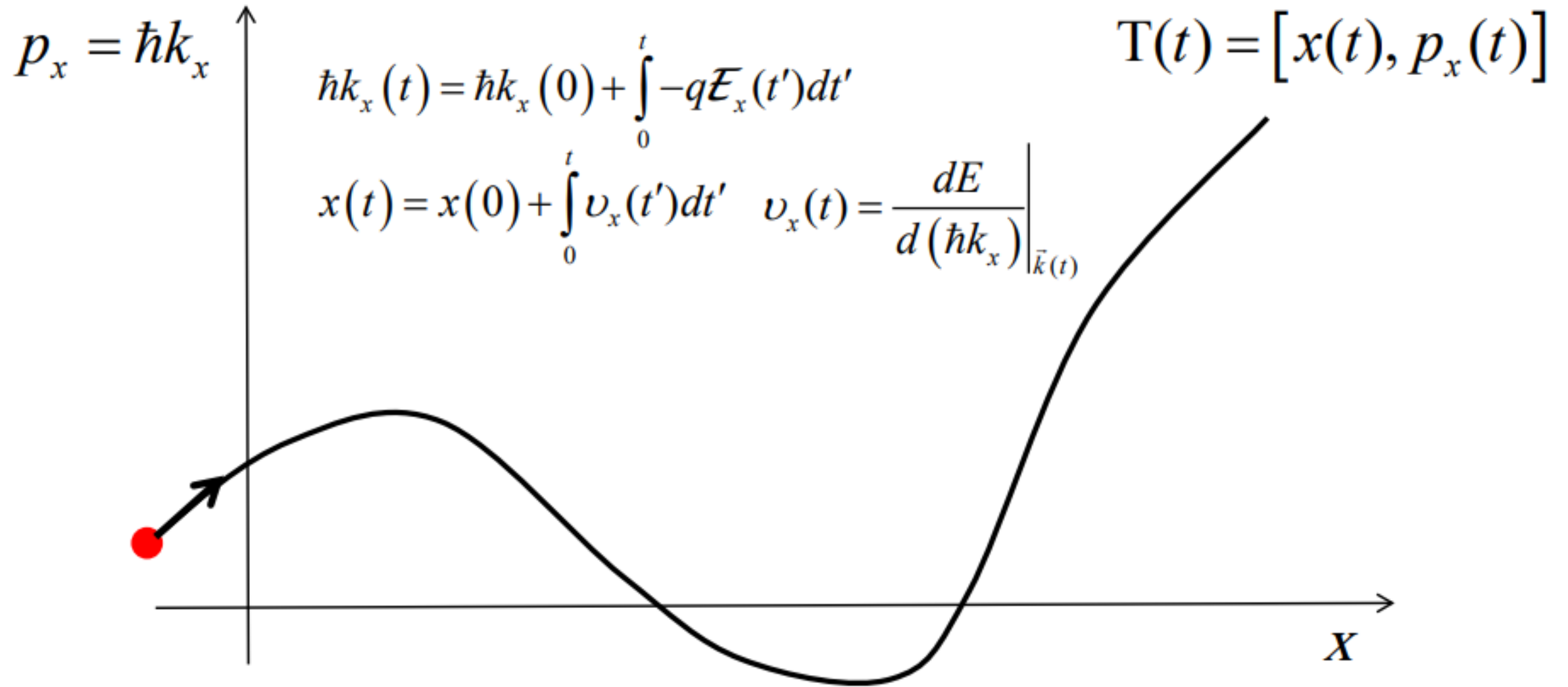
Boltzmann-Transport Equation

- $f(r, k, t)$: The probability that a state with wavevector $\vec{k}$ is occupied by an electron with position $\vec{r}$ at time t
- In equilibrium, $f(r, k, t) = f_0(r, k) = f(E)$

Boltzmann-Transport Equation expresses the total time rate of change of the distribution function, $f(r, k, t)$ which describes the occupancy of allowed energy states involved in transport processes



Trajectories in Phase Space



At time t ; probability distribution function $f = f(x, y, z, p_x, p_y, p_z, t)$

$$\frac{df}{dt} = \left(\frac{\partial f}{\partial x}\right) \frac{dx}{dt} + \left(\frac{\partial f}{\partial y}\right) \frac{dy}{dt} + \left(\frac{\partial f}{\partial z}\right) \frac{dz}{dt} + \left(\frac{\partial f}{\partial p_x}\right) \frac{dp_x}{dt} + \left(\frac{\partial f}{\partial p_y}\right) \frac{dp_y}{dt} + \left(\frac{\partial f}{\partial p_z}\right) \frac{dp_z}{dt} + \left(\frac{\partial f}{\partial t}\right) \frac{dt}{dt}$$

$$\frac{df}{dt} = \left[\frac{\partial f}{\partial x} v_x + \frac{\partial f}{\partial y} v_y + \frac{\partial f}{\partial z} v_z \right] + \left[\frac{\partial f}{\partial p_x} F_x + \frac{\partial f}{\partial p_y} F_y + \frac{\partial f}{\partial p_z} F_z \right] + \left(\frac{\partial f}{\partial t} \right)$$

$$\frac{df}{dt} = \left[\left(\hat{i} \frac{\partial}{\partial x} + \hat{j} \frac{\partial}{\partial y} + \hat{k} \frac{\partial}{\partial z} \right) f \right] \cdot (\hat{i} v_x + \hat{j} v_y + \hat{k} v_z) - \left[\left(\hat{i} \frac{\partial}{\partial p_x} + \hat{j} \frac{\partial}{\partial p_y} + \hat{k} \frac{\partial}{\partial p_z} \right) f \right] \cdot (\hat{i} F_x + \hat{j} F_y + \hat{k} F_z) + \left(\frac{\partial f}{\partial t} \right)$$

$$\frac{df}{dt} = (\vec{\nabla}_r f) \cdot (\vec{v}) + (\vec{\nabla}_p f) \cdot (\vec{F}) + \left(\frac{\partial f}{\partial t} \right)$$

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$$\frac{df}{dt} = \left(\frac{\partial f}{\partial t} \right) + \vec{v} \cdot \vec{\nabla}_r f + \vec{F} \cdot \vec{\nabla}_p f \quad (\text{since } \vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a})$$

➤ In the presence of electric field ($\vec{E}$) and magnetic field ($\vec{B}$);

$$\vec{F} = q[\vec{E} + (\vec{v} \times \vec{B})]$$

$$\frac{df}{dt} = \left(\frac{\partial f}{\partial t} \right) + \vec{v} \cdot \vec{\nabla}_r f + q[\vec{E} + (\vec{v} \times \vec{B})] \cdot \vec{\nabla}_p f$$

Diffusion term: $v \cdot \nabla_r f$

$$\triangleright v \cdot \nabla_r f = v \cdot \frac{\partial f}{\partial r} = v \cdot \frac{\partial f}{\partial T} \frac{\partial T}{\partial r} = v \cdot \frac{\partial f}{\partial T} \nabla_r T$$

$$v = \frac{p}{m} = \frac{\hbar k}{m} \quad \& \quad \varepsilon = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$$\frac{\partial \varepsilon}{\partial k} = \frac{\hbar^2 k}{m} \Rightarrow \frac{1}{\hbar} \frac{\partial \varepsilon}{\partial k} = \frac{\hbar k}{m} = v$$

$$\triangleright v \cdot \nabla_r f = \frac{1}{\hbar} \frac{\partial \varepsilon}{\partial k} \cdot \nabla_r f = \frac{1}{\hbar} \nabla_k \varepsilon \cdot \nabla_r f$$

Drift term: $\frac{1}{\hbar} F \cdot \nabla_k f$

$$\frac{1}{\hbar} F \cdot \nabla_k f = -\frac{e}{\hbar} \left(E + \frac{1}{c} (\mathbf{v} \times \mathbf{B}) \right) \cdot \nabla_k f$$

$$\frac{\partial f}{\partial k} = \frac{\partial f}{\partial \varepsilon} \frac{\partial \varepsilon}{\partial k}$$

$$\mathbf{v} = \frac{\mathbf{p}}{m} = \frac{\hbar \mathbf{k}}{m} \quad \& \quad \varepsilon = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$$\frac{\partial \varepsilon}{\partial k} = \frac{\hbar^2 k}{m} \Rightarrow \frac{1}{\hbar} \frac{\partial \varepsilon}{\partial k} = \frac{\hbar k}{m} = \mathbf{v} \Rightarrow \frac{\partial \varepsilon}{\partial k} = \hbar \mathbf{v}$$

$$\frac{\partial f}{\partial k} = \frac{\partial f}{\partial \varepsilon} \frac{\partial \varepsilon}{\partial k} = \frac{\partial f}{\partial \varepsilon} \hbar v_x$$

Boltzmann transport equation (BTE)

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + v \cdot \nabla_r f + F \cdot \nabla_p f$$

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + v \cdot \nabla_r f + \frac{1}{\hbar} F \cdot \nabla_k f$$

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + v \cdot \nabla_r f - \frac{e}{\hbar} E \cdot \nabla_k f$$

$$\triangleright \frac{df}{dt} = \frac{\partial f}{\partial t} + \boldsymbol{v} \cdot \nabla_r f + \frac{1}{\hbar} \boldsymbol{F} \cdot \nabla_k f$$

At steady state ($\frac{\partial f}{\partial t} = 0$):

$$\frac{df}{dt} = \boldsymbol{v} \cdot \nabla_r f + \frac{1}{\hbar} \boldsymbol{F} \cdot \nabla_k f$$

$$\frac{df}{dt} = \left(\frac{\partial f}{\partial t} \right)_{scattering}$$

$$\text{Boltzmann Transport Equation: } - \left(\frac{\partial f}{\partial t} \right)_{scattering} + \boldsymbol{v} \cdot \nabla_r f + \frac{1}{\hbar} \boldsymbol{F} \cdot \nabla_k f = 0$$

$$-\left(\frac{\partial f}{\partial t}\right)_{scattering} + v \cdot \nabla_r f + \frac{1}{\hbar} F \cdot \nabla_k f = 0$$

Can be solved if F and $\left(\frac{\partial f}{\partial t}\right)_{scattering}$ are known.

- $\left(\frac{\partial f}{\partial t}\right)_{scattering}$: accounts for local changes in distribution at $\vec{r}$ due to scattering, generation-recombination...
- $v \cdot \nabla_r f = \left(\frac{\partial f}{\partial t}\right)_{diff}$: accounts for changes in distribution due to concentration gradient (diffusion).
Particle at $\vec{r}$ moves $\vec{r}'$
- $\frac{1}{\hbar} F \cdot \nabla_k f = \left(\frac{\partial f}{\partial t}\right)_{force}$: accounts to changes in distribution due to forces. Particle in state $\vec{k}$ changes to state $\vec{k}'$

$I_k(f)$ = Collision integral = $\left(\frac{\partial f}{\partial t}\right)_{scattering}$: Contribution from all the scattering process.

Models of Scattering

- Electron-Electron scattering
 - Electron-Phonon scattering
 - Electron-impurity scattering
 - Electron-photon scattering
 - Electron-magnon scattering
 - Electron-plasmon scattering
- Simple model of collision integral (Relaxation time approximation)

Boltzmann Transport Equation in the steady state: $-\left(\frac{\partial f}{\partial t}\right)_{scattering} + v \cdot \nabla_r f + \frac{1}{\hbar} F \cdot \nabla_k f = 0$

- The action of applied force and gradient tend to disturb the distribution f from its equilibrium value f_0 . If this disturbance is removed, the scattering process will tend to restore equilibrium.
- When the change in the distribution is not large compared to its initial value,

$$\left(\frac{\partial f(k)}{\partial t}\right)_{scattering} = \frac{f(k) - f_0(k)}{\tau_m} \quad \text{Relaxation time approximation}$$

;where the constant of proportionality τ_m is the (momentum) relaxation time.

- In equilibrium,

$$\begin{aligned} \left(\frac{\partial f(k)}{\partial t}\right)_{scattering} &= \frac{f(k) - f_0(k)}{\tau_m} = 0 \\ \Rightarrow f(k) &= f_0(k) \end{aligned}$$

- In general, τ_m depends on the electron energy and is different for **different scattering mechanism**.

Thank You